

HOPPER INFORMATION SERIES

03 / BLOCK CODING

Coding blocks reference

What each Hopper Studio block category does, what it returns and how commands sequence.

FLIGHT

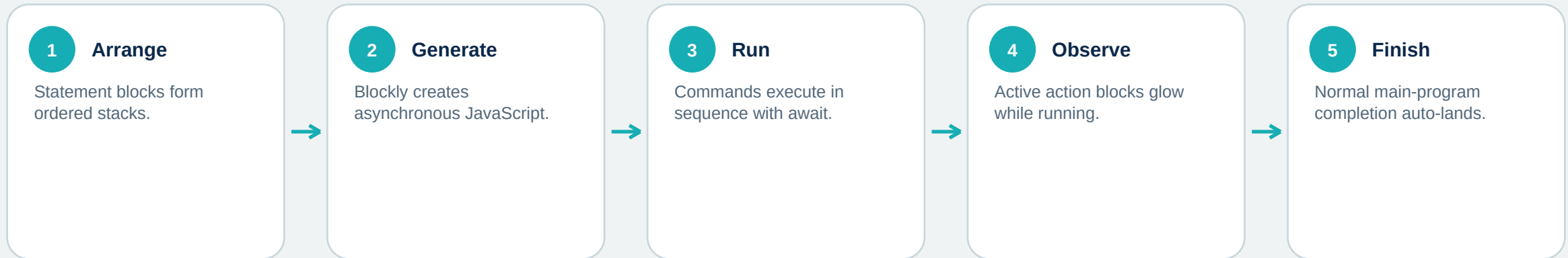
VISION

LOGIC + LOOPS





How a block program becomes a flight



when program starts

take off

fly forward for 1 sec at 15%

land

GENERATED JAVASCRIPT

```
1  await runtime.runBlock(...
2    await drone.takeOff();
3  );
4  await drone.fly("forward", 1, 15);
5  await drone.land();
```

Structural blocks may not glow; concrete flight, wait, accessory and vision actions do.



Start & Events

when program starts

Container for the main stack. Other top-level stacks can also be generated.

stop program

Ends execution but does NOT send a land command. Land first or use UI STOP & LAND.

wait [seconds]

Pauses in sequence. Negative/coercion failures become zero.

print [value]

Writes a line to the in-app program console.

when [key] is [pressed/released]

Arrow, Space and a-z events. Event programs keep listening until Stop.

[key] key is pressed

Boolean from the live pressed-key set.

continue if [condition]

False returns from the current main, event or function.

SAFETY

The red UI STOP & LAND is the dependable way to cancel tasks and send a landing command.



Mini Drone: flight blocks

take off / land / hover

Awaited state-changing actions.

rotate [degrees] [direction]

Timed/open-loop yaw at about 180 deg/s, then settle.

set [axis] to [power]%

Persistent pitch/roll/yaw/gaz power until reset. 'Altitude' means gaz, not height.

center on AprilTag

Repeated scans, translation pulses and yaw alignment; does not control height or land.

fly [direction] for [s] at [power]%

6 directions; time ≥ 0 ; teach 0-100% power; default 1 s / 15%.

flip [direction]

Forward/back/left/right; preserve safe height and clear space.

reset movement

Zeros all motion axes; does not land.

cut off motors

Emergency only. Immediate; block/code has no confirmation.

Fly adds a stabilization wait after motion. set-axis does not.



Mini Drone: state, events and accessories

SENSORS + EVENTS

- battery level -> number 0-100 or null before telemetry.
- drone is flying / landed -> Boolean.
- wait until battery changes -> asynchronous wait.
- when drone starts flying / lands / crashes / battery changes.

ACCESSORIES

- take and store photo -> session gallery JPEG from camera pixels.
- open / close grabber -> needs physical claw; simulator is a timing stub.
- fire cannon -> needs physical cannon; simulator is a timing stub.

NOT AVAILABLE AS BLOCK VALUES

GPS | x/y room position | real altitude | velocity | attitude | heading | raw IMU | range

'set altitude' is signed vertical motor power. It is not an altitude setpoint.



Camera Vision blocks

BINARY

camera sees binary white/black at threshold + average

camera sees binary white/black at center pixel

OBJECTS

scan for objects

camera sees [label] at [confidence]%

x/y coordinate of [label] at [confidence]%

CUSTOM

custom model sees [label] at [confidence]%

APRILTAGS

scan for april tags

camera sees april tag with ID [any/0-586]

center on AprilTag (in Flight category)

Fresh scan: binary predicates, sees object, custom sees and sees AprilTag.

Saved state only: object coordinate. 0 can mean centered OR not previously detected.



Logic, loops, math, variables and functions

LOGIC

- if / else
- comparisons
- and / or / not
- true / false
- ternary

LOOPS

- forever
- for N seconds
- repeat count
- while / until
- counted for
- break / continue

MATH

- arithmetic
- unary math
- trig
- rounding
- modulo
- random

VARIABLES

- create
- get
- set
- change

FUNCTIONS

- define
- parameters
- return value
- awaited calls

Loops yield so Stop can interrupt. Generated custom functions are async; calls are awaited.



Build one intentional, safe mission

when program starts

take off

repeat for 10 seconds

fly forward for 1 sec at 15%

if camera sees white paper

land

land

MISSION RULES

- Place all intended executable stacks deliberately; detached top-level stacks may still run.
- Use one start hat for the main mission.
- End with land even when a condition is never met.
- Use the UI STOP & LAND for intervention.

SOURCES AND FURTHER READING

Hopper Studio block definitions and JavaScript generators

lib/blockly.ts in the Hopper Studio project

Hopper Studio execution and safety behavior

components/HopperStudio.tsx, lib/runtime.ts and README.md